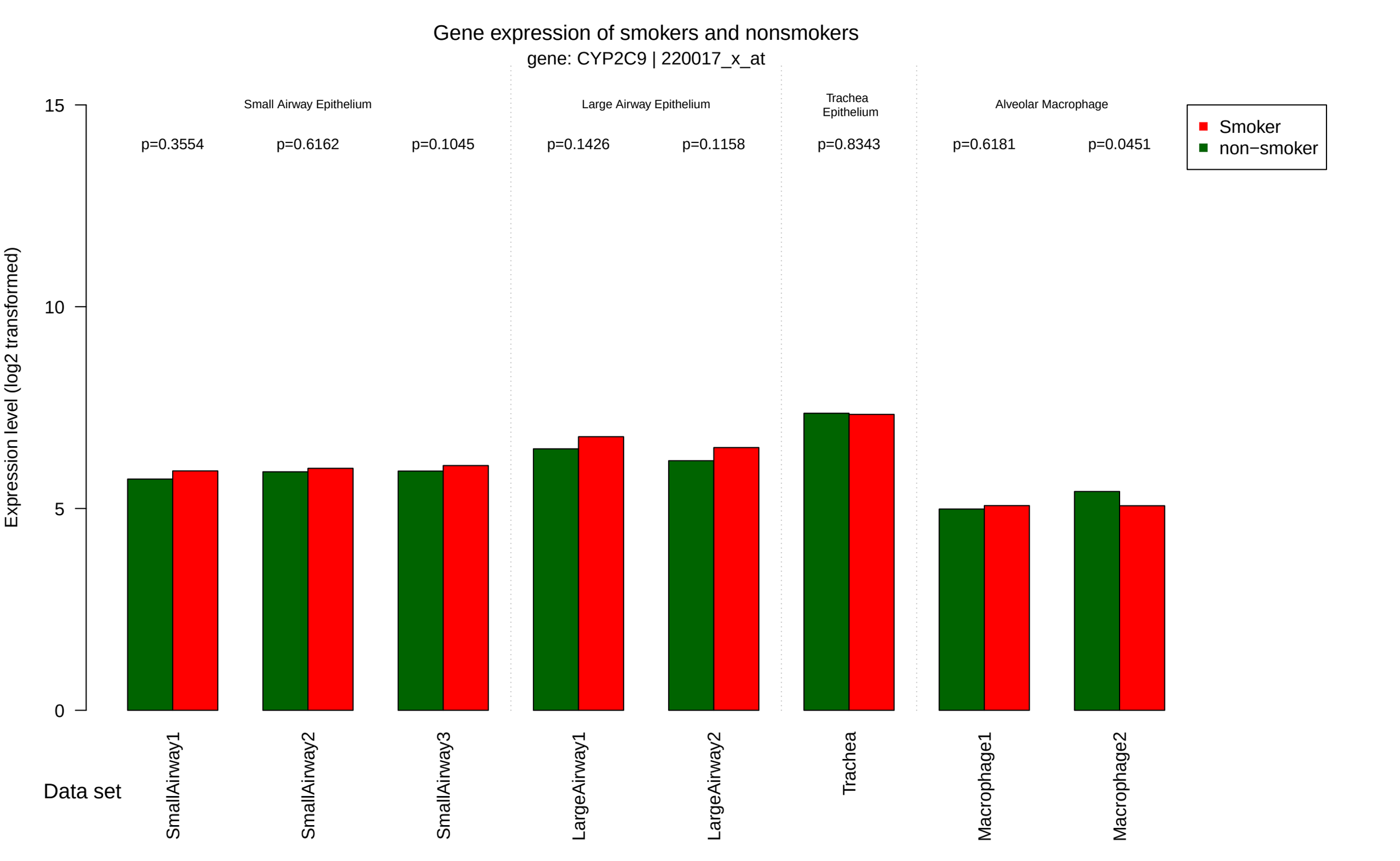
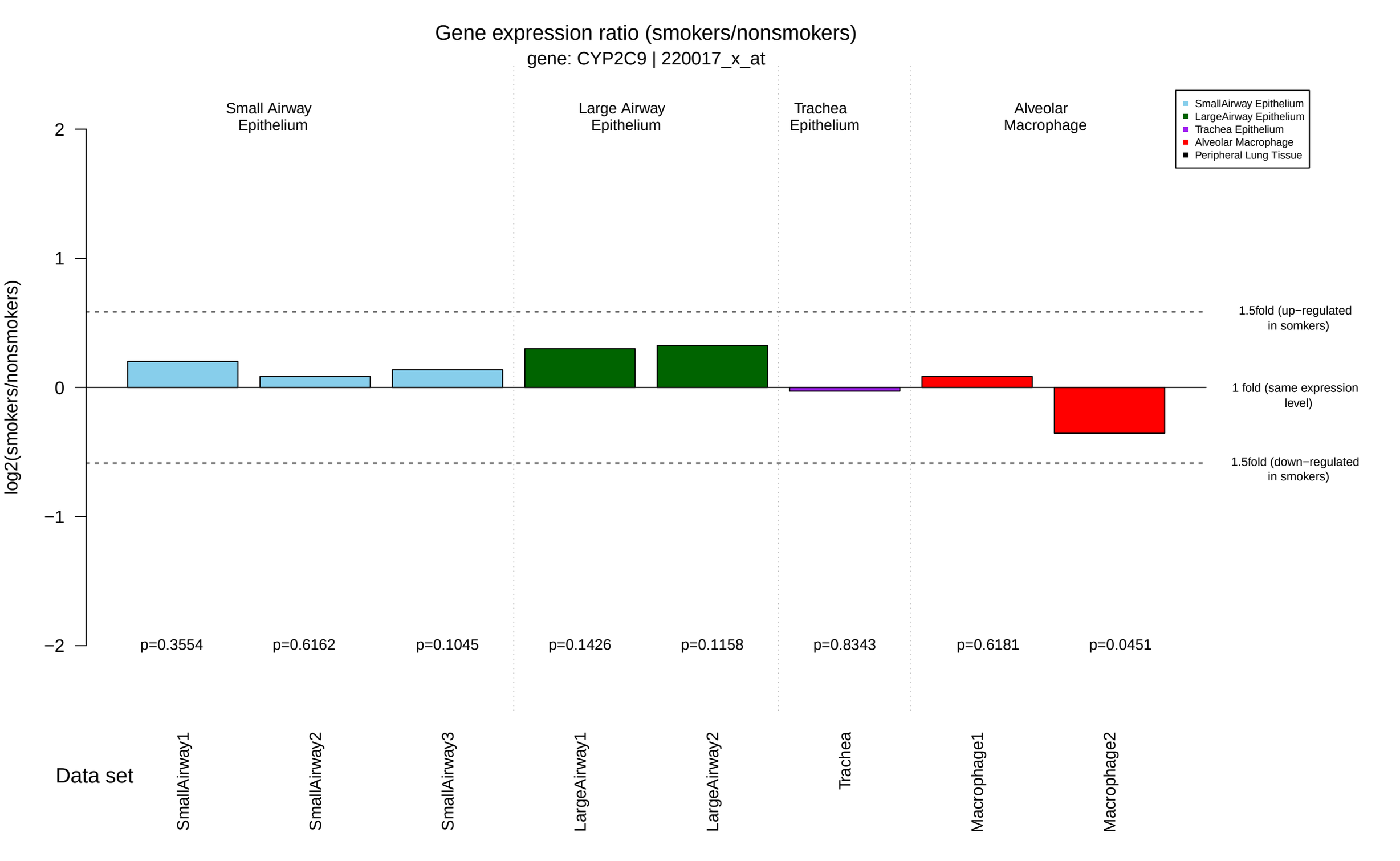
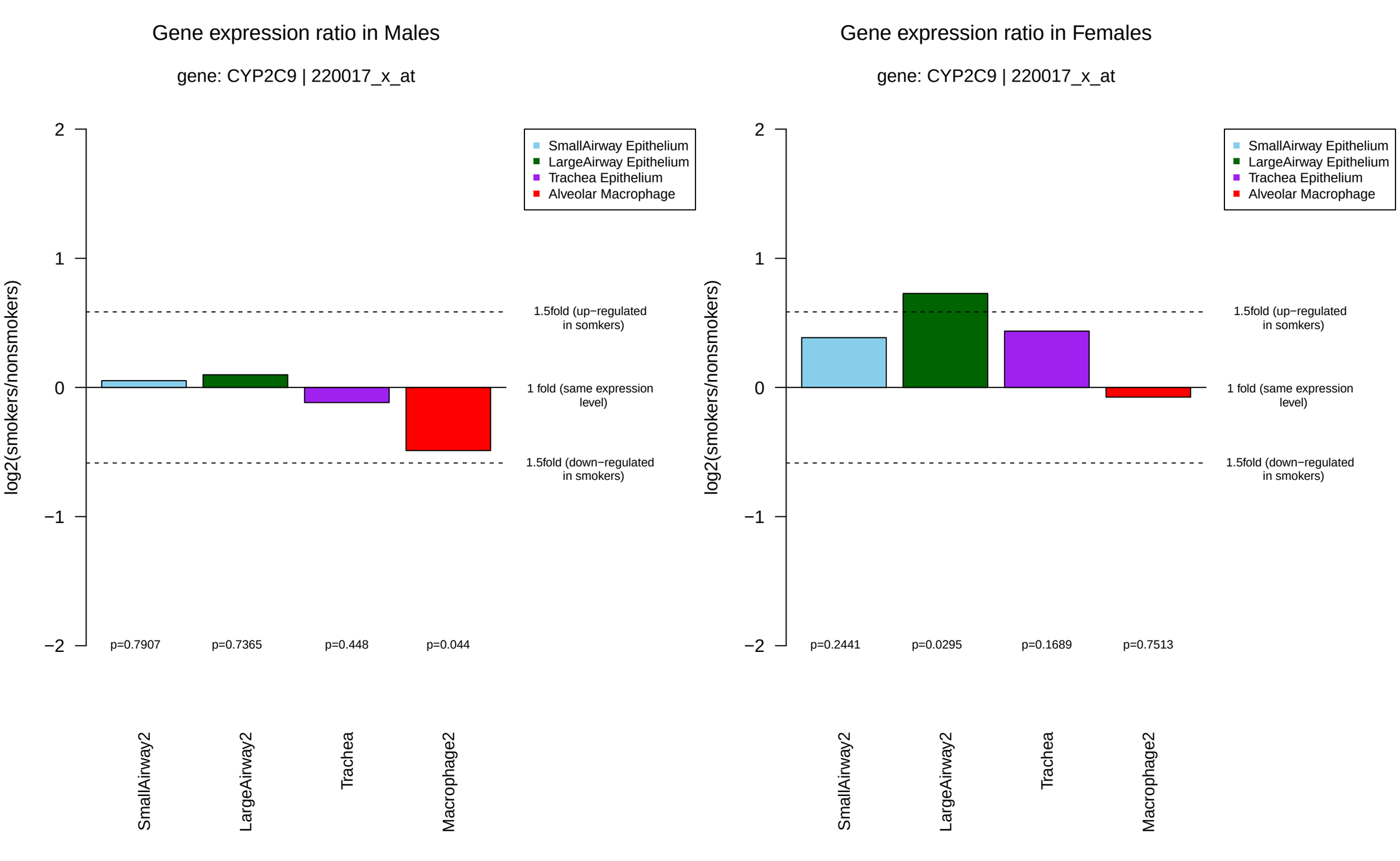
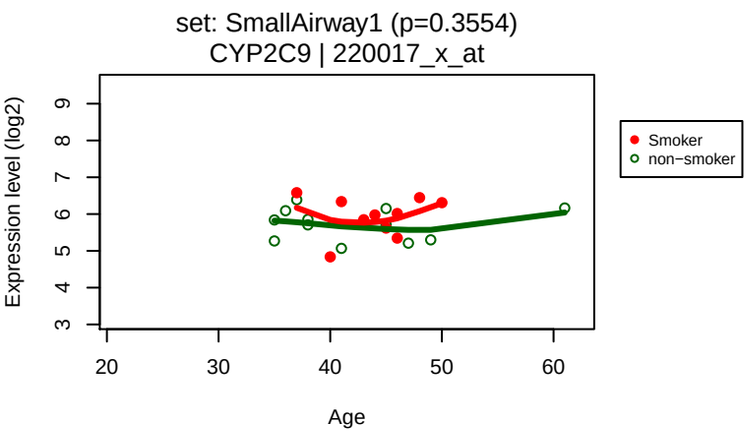
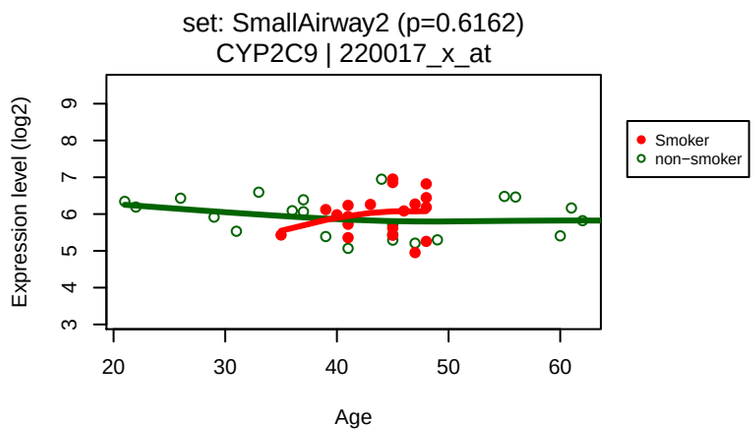
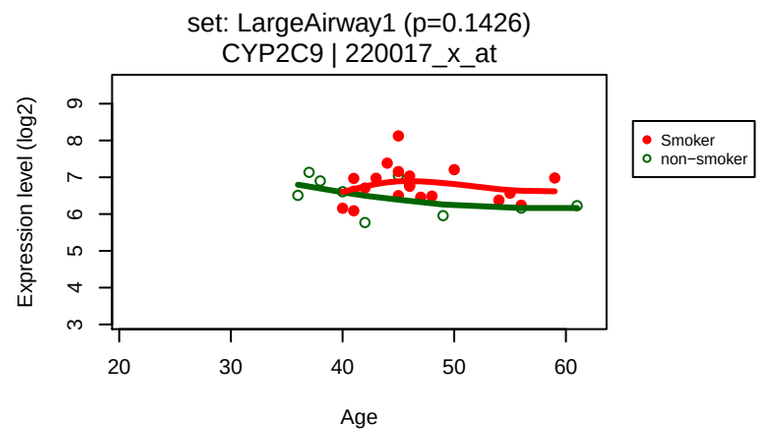

Gene expression of smokers and nonsmokers
gene: CYP2C9 | 220017_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3554
p=0.6162
p=0.1045
p=0.1426
p=0.1158
p=0.8343
p=0.6181
p=0.0451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

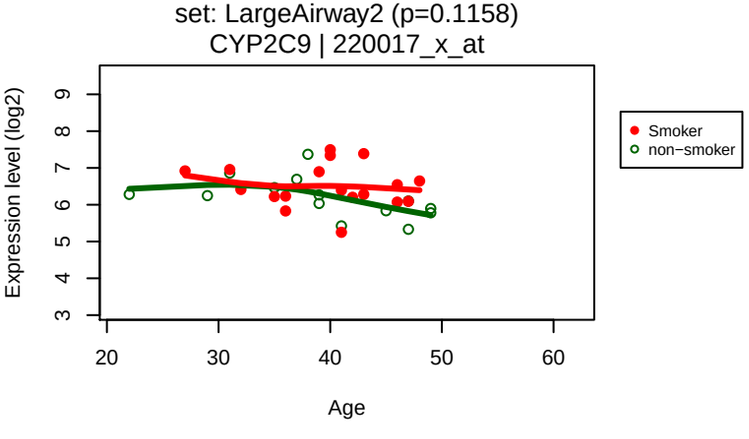

Gene expression ratio (smokers/nonsmokers)
gene: CYP2C9 | 220017_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3554
p=0.6162
p=0.1045
p=0.1426
p=0.1158
p=0.8343
p=0.6181
p=0.0451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

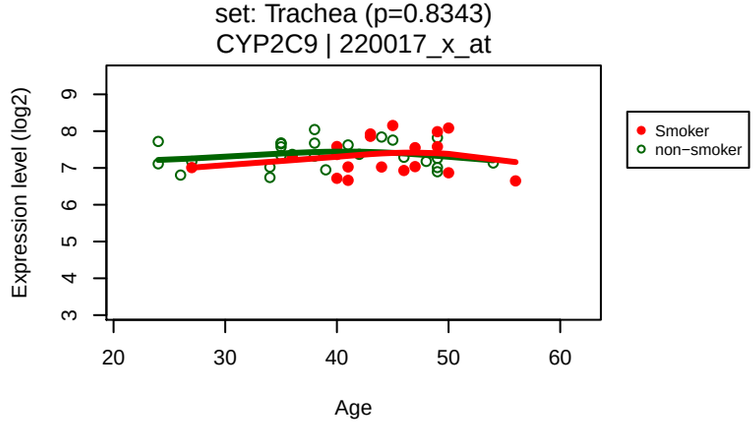

Gene expression ratio in Males
gene: CYP2C9 | 220017_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7907
p=0.7365
p=0.448
p=0.044
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2C9 | 220017_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2441
p=0.0295
p=0.1689
p=0.7513
SmallAirway2
LargeAirway2
Trachea
Macrophage2

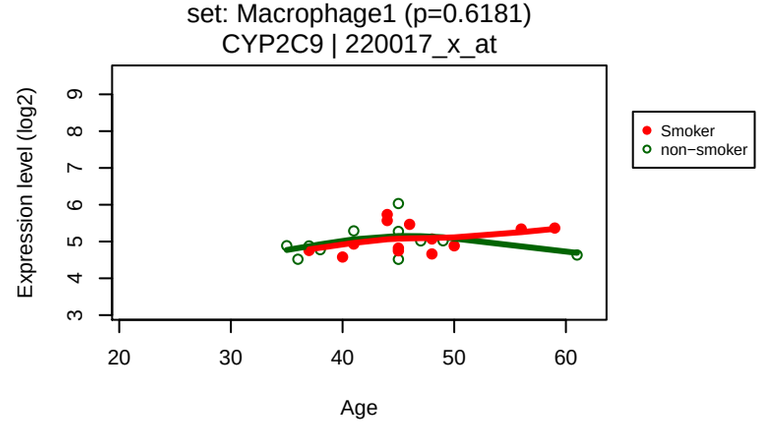

set: SmallAirway1 (p=0.3554)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker

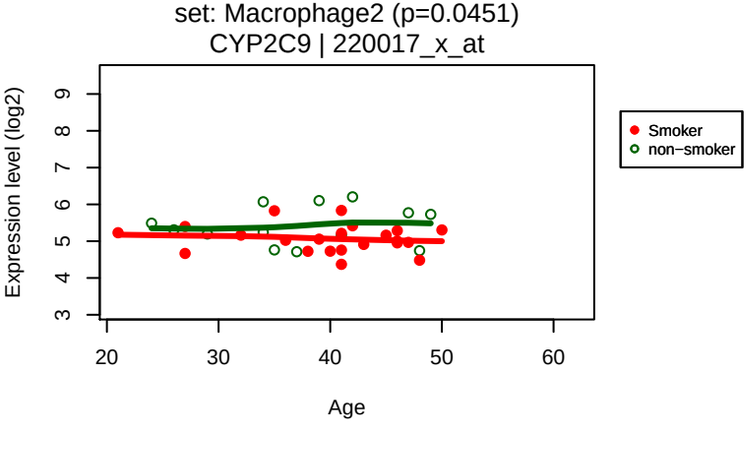

set: SmallAirway2 (p=0.6162)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker

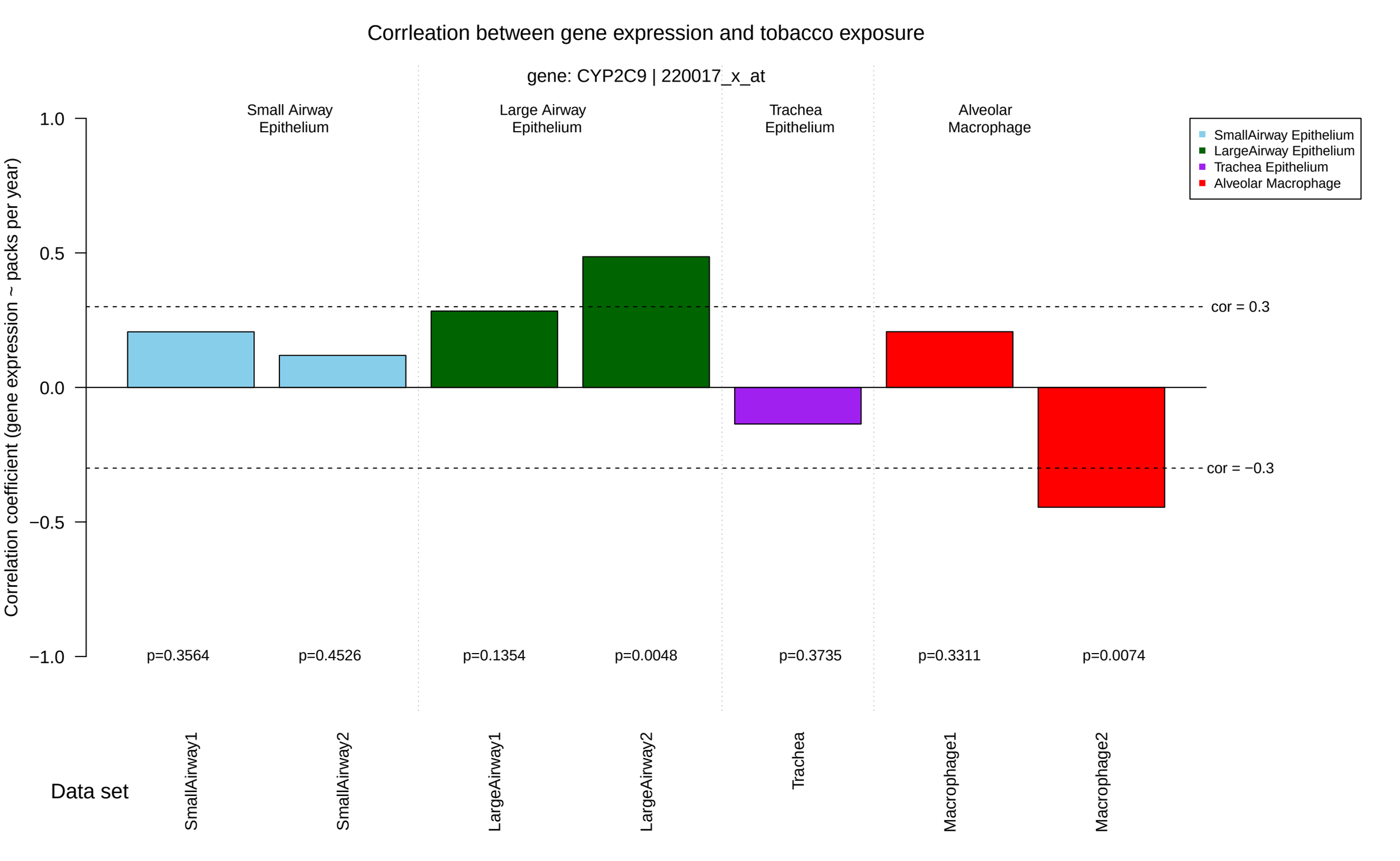

set: LargeAirway1 (p=0.1426)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1158)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8343)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6181)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0451)
CYP2C9 | 220017_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2C9 | 220017_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3564
p=0.4526
p=0.1354
p=0.0048
p=0.3735
p=0.3311
p=0.0074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2